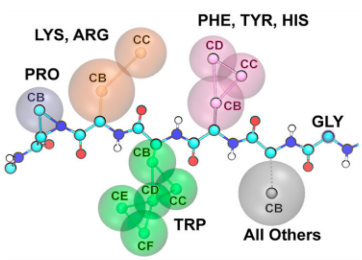
Shanlong Li | Research Summary

My research focuses on developing multi-scale computational methods to advance our understanding of the dynamics and interactions of bio- and designed macromolecules, including recognition, self-assembly, phase transitions, and aggregation. I have made significant contributions to studies of the condensation of intrinsically disordered proteins (IDPs) and flexible RNAs, as well as the self-assembly of amphiphilic alternating copolymers. Critically, my work involves close collaboration with experimentalists, integrating simulation and experiment to reveal and interpret complex behaviors that emerge in biological and soft matter systems. Major contributions from my PhD and postdoctoral works are summarized below (References are listed in the figure captions).

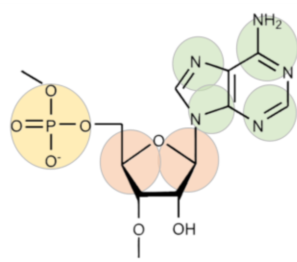
I. Efficient models for biomolecules and polymers. A primary focus of my research is to develop accurate and effective coarse-grained (CG) models for simulating self-assembly and phase behaviors of (bio)macromolecules occurring on large time and length scales. By careful parameterization based on experimental data, all-atom simulations, and quantum calculations, I have developed several CG models with unprecedented accuracy, including intermediate resolution models for protein[1], RNA[2], and polymers[3, 4], and mesoscale polymer models[6-8]. These models have played key roles in understanding and interpreting complex experimental observations and in guiding the design of new biomolecular and polymer materials. All the models have been distributed in the widely available OpenMM, Gromacs, and Hoomd-blue packages with GPU acceleration.

Protein Model



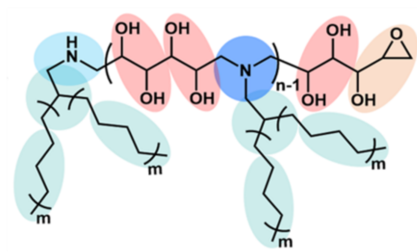
A hybrid resolution model for protein condensates (***1.*** *JACS 2024, 146, 342*)

RNA Model



A sub-residual model for RNA condensates (***2.*** *PNAS 2025, 122, e2504583122.*)

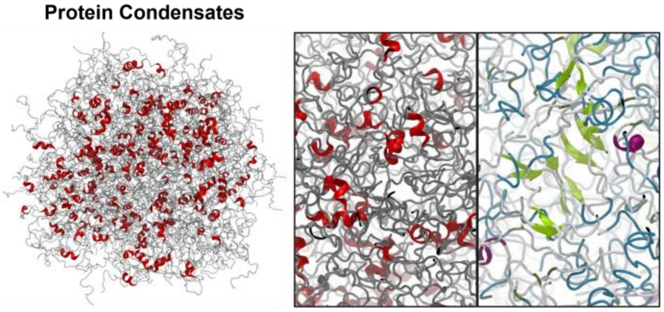
Polymer Model



Several CG models for polymer self-assembly (***3.*** *JPCB 2022, 126, 1830;* ***4.*** *PCCP 2020, 22, 5577*)

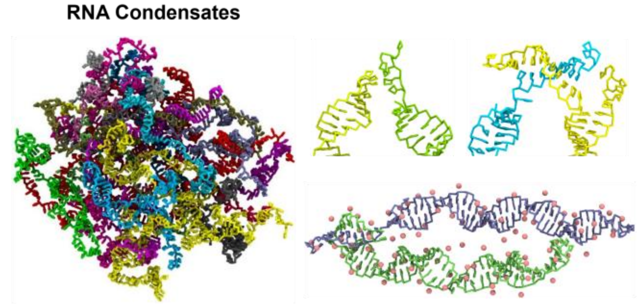
II. Biomolecular condensation and material properties. Biomolecular condensates play a central role in diverse cellular functions and human diseases. Using CG protein and RNA models, my work has revealed the coupling between protein secondary structure and phase separation, which has greatly promoted the understanding of the relationship among protein structure, phase separation, gelation, and solidification, and the contribution of protein backbones to phase separation[1, 5]. In addition, I have also revealed the roles of intramolecular folding and intermolecular pairing in the condensation, percolation, and gelation of RNA[2]. My latest work further revealed how competition of homotypic and heterotypic interactions drives reentrance phase transitions of protein/RNA and determines material properties.

Protein Condensates



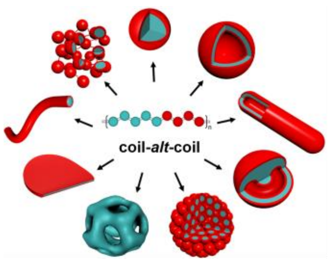
Coupling between phase separation and secondary structures of IDPs (***1.*** *JACS 2024, 146, 342;* ***5.*** *Biochem. Soc. Trans., 2024, 52, 319*)

RNA Condensates

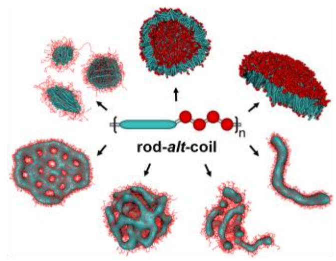


Interplay between self-folding and intermolecular paring in RNA condensates (***2.*** *PNAS 2025, 122, e2504583122*)

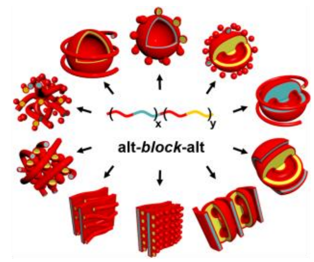
III. Self-assembly of alternating copolymers. Amphiphilic alternating copolymers represent promising candidates for the development of advanced smart nanomaterials through phase separation and self-assembly. However, the underlying molecular mechanisms driving these processes remain poorly understood. To address this gap, I have established a comprehensive framework for simulating the phase behaviors of coil-coil, coil-rod, and multidomain alternating copolymers, offering novel insights into these fundamental properties[6, 7]. In particular, our works highlight the morphological diversity of self-assembled structures, identify their unique characteristics, and explore potential applications in smart nanomaterials[4, 8]. Several of these findings have been validated experimentally, underscoring the great potential of multiscale computational simulations for advancing smart nanomaterial design.



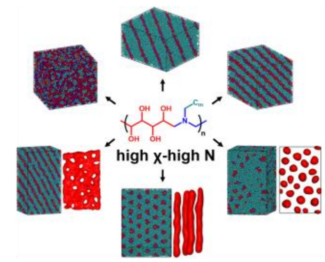
Self-assembly of alternating copolymers with two coil segments (***6.*** *Sci. China Chem. 2019, 62, 226*)



Self-assembly of alternating copolymers with rigid and coil segments (***7.*** *PCCP 2019, 21, 25148*)

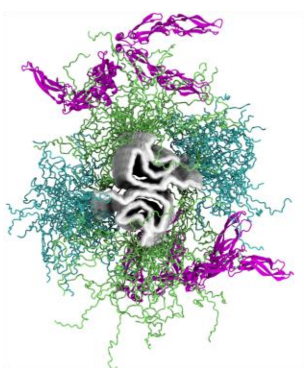


Design blocked alternating copolymers for complex nanoparticles (***8.*** *JPCB 2019, 123, 8333*)

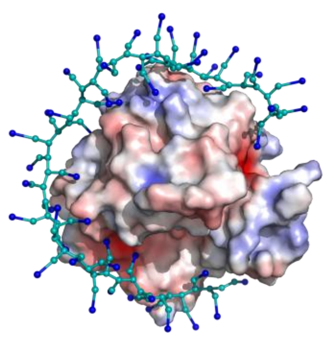


Design high-χ alternating copolymers for sub-5 nm nano-patterns (***4.*** *PCCP 2020, 22, 5577*)

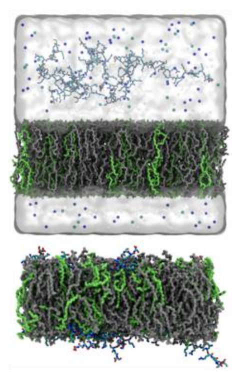
Experimental collaboration. A hallmark of my PhD and postdoctoral research is extensive collaboration with experimentalists, where I strive to provide molecular-level insights into complex experimental phenomena and make testable predictions. These collaborations have spanned a wide range of topics—from molecular-level investigations of smart materials, such as the packing of porphyrin groups in polymer vesicles[9], to microscale studies of molecular interactions, including protein–polymer and peptide–bilayer binding, and mesoscale analyses of phase behavior[10, 11], such as polymer vesicle fusion and photo-induced fission. Through an integrative computational and experimental approach, we have been able to make important contributions in all these endeavors.



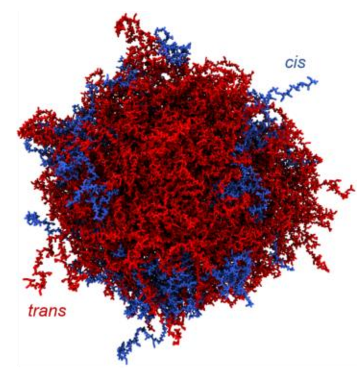
protein-protein binding



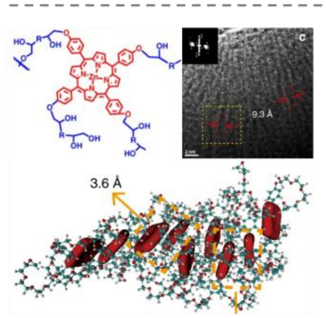
protein-polymer binding



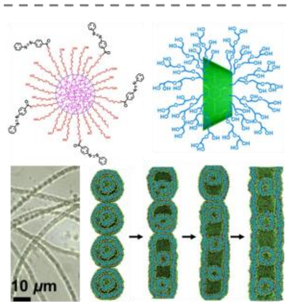
protein-bilayer binding



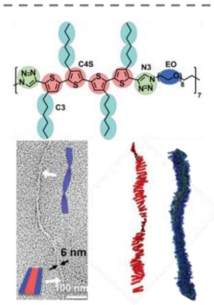
peptide confirmation in condensates



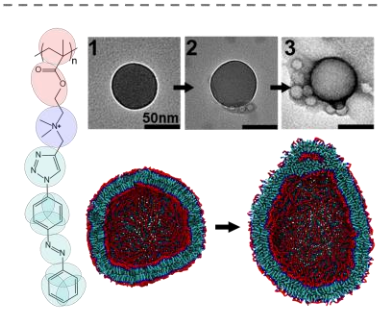
porphyrin arrangement (***9.*** *Nat Commun 2020, 11, 1724*)



hierarchical self-assembly (***10.*** *Angew. Chem. Int. Ed. 2022, 61, e202213178*)



nanosheet formation (***11.*** *Macromol. Rapid Commun. 2022, 43, 2200437*)



polymersome photo-fission